Reg. No. | | | | | | | | | | |

[16MSCCH402]

FOURTH SEMESTER UNIVERSITY EXAMINATION, MAY - 2019

SPECTROSCOPY - II

Time: 03 Hours **Maximum Marks:70**

Instructions: **I. Question 1 is compulsory, carries 10 marks.**
II. Questions 2 to 7 carry 15 marks each, answer any four.

1. Answer any FIVE **5×2=10**

a. What does it mean by fingerprint region of an IR spectrum?

b. Give a graphic description of electronic transition with increasing order of energy in a UV- spectrum.

c. When can a molecule be assigned as EPR active?

d. How for a switchable molecule spin cross over effect work?

e. Describe temperature induced spin-state switching.

f. Describe the principle photoelectron spectroscopy with a neat diagram.

g. How are X-rays produced using different sources?

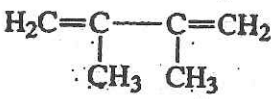
2. a. Describe the frequency changes in the IR spectrum of endocyclic and exocyclic compounds having different ring size distribution of double bonds.

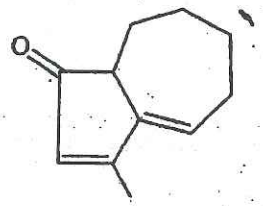
b. Explain the effect of aromatic substitution with relation to bending vibrations?

(10+5)

3. a. How solvents affect the λ_{max} value in UV spectroscopy and also describe the probability of transitions in UV spectroscopy.

b. Explain how Woodward-Fieser rules can apply to calculate the λ_{max} value for the following :

i. 

ii. 

(10+5)

4. a. i. With a magnetic field of 22000 Gauss, of g = 2 and $\beta = 9.273 \times 10^{-24}\ JT^{-1}$, estimate the ESR frequency.

ii. Differentiate the ESR and NMR spectroscopic methods with a note on the limitation of ESR.

b. Why one among the following is ESR active?

i. O_2 or H_2

ii. NO or DPPH

10+5

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5. a. Describe the nuclear quadrupole electric interaction in NQR spectroscopy.

b. Describe the difference in the Mossbauer spectrum of Prussian blue and Turnbulls blue.

c. Explain the basic principle involved in NQR and list out its applications.

5+5+5

6. a. What principles of electron spectroscopy apply for chemical analysis? Describe the details of instrumentation.

b. Explain the phenomenon associated with photoelectron spectroscopy. **10+5**

7. a. Describe the technique used for surface topology measurement with its principle and instrumentation.

b. What are the applications of atomic force spectroscopy? **10+5**

$F \propto x$

$\boxed{F = kx}$